Type 2: Application Project

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DEEMED TO BE UNIVERSITY UNDER SECTION 3 OF UGC ACT, 1956

Intelligent Nutrient Management for Aeroponics

23CSE498 – Project Phase 2 (Panel Review 1) | **Team Number: 19**

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4	CB.SC.U4CSE23239	Karthik Vishal G
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Guide: **Dr. Vishnu S**, Assistant Professor (Sr.Gr), Department of CSE

September 6, 2026

1. Recap of Aeroponics

Plants Grow Without Soil

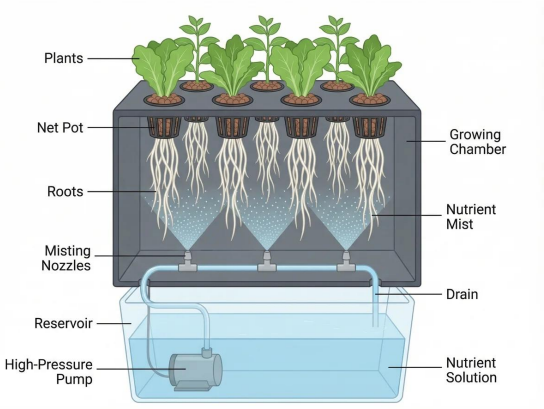
- Roots remain suspended in a controlled chamber, receiving abundant oxygen.

Nutrient-Rich Mist

- A fine mist supplies water and essential nutrients directly to the roots while using less water.

Efficient and Faster Growth

- Direct access to oxygen, water, and nutrients supports faster growth while saving space and water.

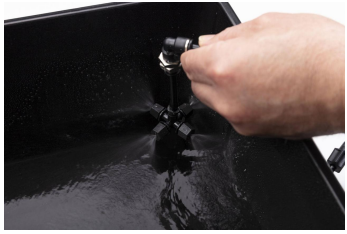


2. Problems Faced in Aeroponics

Major issues faced in aeroponics

- Water and nutrient buffering problems.
- Pump failure and dependence on electrical power.
- Uneven nutrient-solution distribution.
- Atomization and nozzle clogging.
- Need for precise misting-system control.
- Rapid plant damage or death when nutrient delivery fails.

Adapted from aeroponics system reviews [1, 2].



Nozzle clogging



Adaptive misting

3. From Problem Exploration to Solution

EC & pH Limitation

- EC and pH show overall solution characteristics.
- They cannot identify individual ions or deficiencies.
- Individual nutrient sensing is needed for precise control.

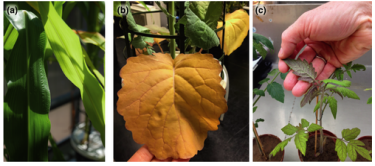
Plant-Based Detection

- Deficiencies produce characteristic visible symptoms.
- Proximal imaging captures these changes.
- Machine learning classifies the deficiency.

Related ML-based detection approach [3].



EC/pH measurement



Nitrogen deficiency



Phosphorus deficiency

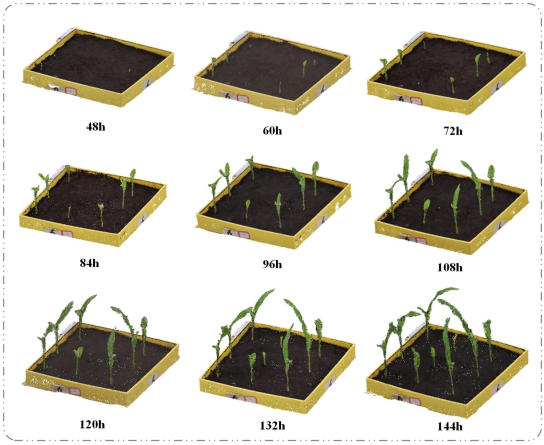
4. Problem Statement

- EC and pH monitoring cannot identify individual nutrient deficiencies; visible symptoms still require expert inspection.
- Fixed nutrient replacement wastes solution when only one nutrient is deficient.
- A non-invasive vision system is needed to detect deficiencies from plant images.
- The detected deficiency should drive nutrient correction and adaptive misting, focusing on calcium nitrate, potassium sulphate, and ammonium phosphate.

5. Experiment

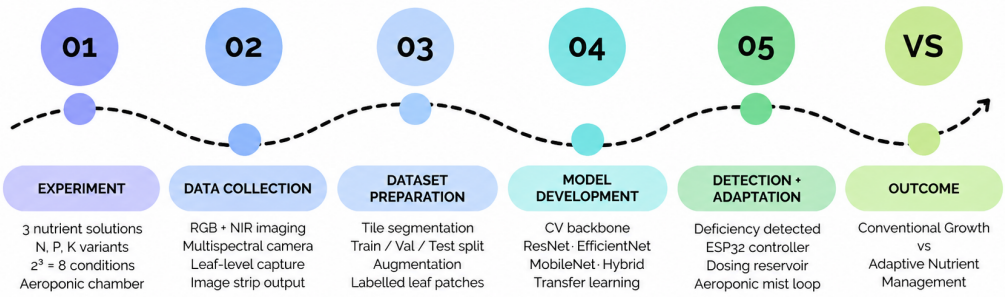
Experimental Design

- **3 Nutrients:** Calcium nitrate, potassium sulphate, and ammonium phosphate.
- $2^3 = 8$ **Combinations:** $000 \rightarrow 001 \rightarrow 010 \rightarrow \cdots \rightarrow 111$.
- **Controlled Growth:** Same seed, environment, and aeroponic conditions.
- **Time-Series Imaging:** Plant images captured throughout growth.
- **Test Set:** Separate plants used for model evaluation.



Design informed by plant phenotyping research [4].

6. Research and Development Pipeline

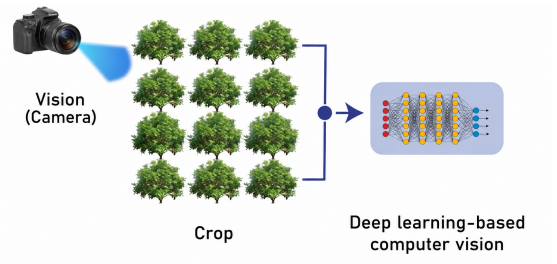


7. Application Architecture & Workflow Design

Rubric: Workflow, Module Interaction, UI Flow, DB Schema, APIs & Deployment (5 Marks)

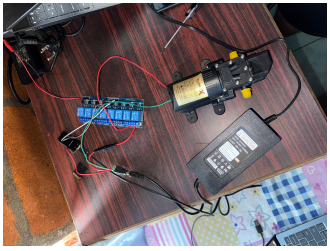
Vision-Based Architecture

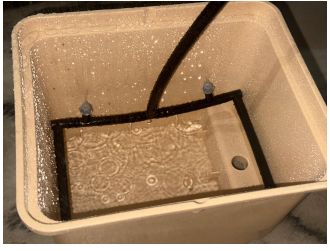
- **Image Acquisition:** Camera captures plant images at defined intervals.
- **AI Processing:** Image preprocessing → nutrient-deficiency classification.
- **Decision & Control:** Identified deficiency → required nutrient correction.
- **Edge Execution:** Processing and control run locally on an edge device such as a Raspberry Pi or ESP32.



8. Test Bed, Hardware Demo & Tech Stack

Rubric: In-depth Understanding of Architecture, Implementation & Tech Stack (5 Marks)





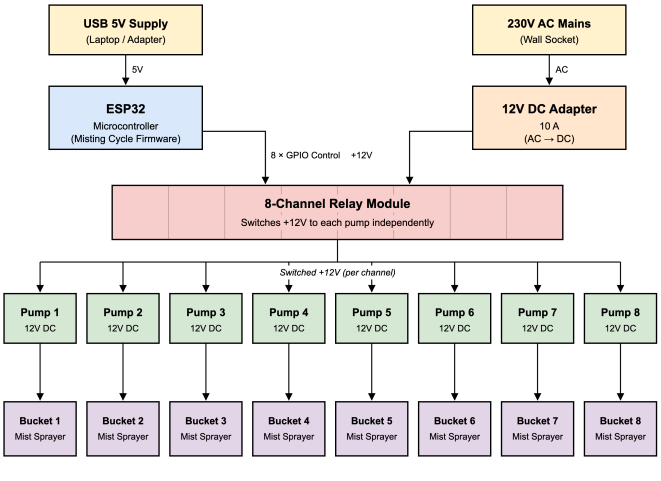
Hardware Stack

- **Controller:** ESP32
- **Switching:** 8-channel relay module
- **Power:** 12V 10A DC adapter
- **Actuators:** 8× 12V diaphragm pump, 125 PSI
- **Setup:** 8 Dutch buckets with mist sprayers
- **Control:** Misting cycle driven by ESP32 firmware

▶ **Watch hardware demo video**

9. Block Diagram (Automated)

Fig. 1 — Block Diagram of the Aeroponic Control System



10. Implementation Progress & Module Functionality

Rubric: Major Modules Implemented, Integrated & Demonstrated (8 Marks)

Module / Feature	Current implementation	Status
Problem Formulation	Research gap and proposed approach established	Completed
Aeroponic Hardware	Chamber, reservoir, pump, misting, and camera setup	Functional
Automation & Control	ESP32 with MOSFET/relay-based misting control	Functional
Nutrient Experiment & Dataset	2^3 combinations and time-series image collection	In Progress
Research Paper Documentation	Literature review, methodology, and experiment documentation	In Progress
Image Analysis & ML	Traditional CV, ResNet, EfficientNet, MobileNet, and hybrid models	Yet to Do
Deficiency Detection	Identify deficient nutrient and integrate correction	Yet to Do

Demonstration: Live hardware demo of the aeroponic setup and automated misting.

Repository: AKASA-Lab GitHub

11. Technical Enrichment Activity Choices

Professional Online Certification Details (Individual Mandatory: 10–20 Hours)

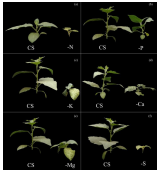
- **Individual MOOC / Online Certification Selection:**

S.No	Student Name	Chosen MOOC / Course Title	Platform	Hours
1	Karthik	Machine Learning with Python	Coursera – IBM	20 Hrs
2	Shibi	Introduction to Computer Vision and Image Processing	Coursera – IBM	20 Hrs
3	Arya	Computer Vision Basics	Coursera – University at Buffalo	12 Hrs
4	Adwaith	Image and Video Processing: From Mars to Hollywood with a Stop at the Hospital	Coursera – Duke University	20 Hrs
5	Abhinav	AWS SimuLearn: Data Analytics Learning Plan	AWS	23 Hrs

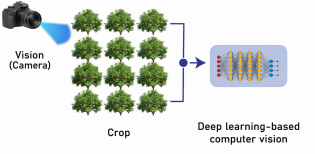
- **Requirement:** Each student must complete one approved 10–20 hour online certification from a recognized platform (Coursera, NPTEL, edX, etc.) relevant to the project.

12. End Result and Deliverables

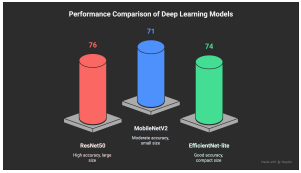
1. Nutrient-Deficiency Dataset



2. Vision-Based Detection



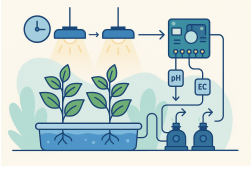
3. Best-Performing ML Model



4. Automated Nutrient Correction



5. Adaptive Aeroponic Management



6. Conventional Cultivation Comparison



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- N. Ganesh, R. Shankar, M. Mahdal, J. S. Murugan, J. S. Chohan, and K. Kalita, “Exploring deep learning methods for computer vision applications across multiple sectors: Challenges and future trends,” *Computer Modeling in Engineering & Sciences*, vol. 139, no. 1, pp. 103–141, 2024. [Online]. Available: http://www.techscience.com/CMES/v139n1/55098

Thank You

Individual Contributions

Team Member	Primary Contribution
Abhinav	Nutrient research, formulation study, and selection of nutrient components.
Adwaith	Literature review, experimental planning, and selection of nutrient and hardware components.
Arya	Research-paper documentation and experimental support.
Shibi	Aeroponic hardware setup and physical system integration.
Karthik	ESP32-based automation, misting control, and hardware-control integration.